

Physical Pharmacy – Midterm Question Bank

Lecture 1 – State of Matter (25 Questions)

Part A: MCQ (10 Questions)

1. Which state of matter has fixed shape and fixed volume?
 - A) Liquid
 - B) Gas
 - C) Solid
 - D) Plasma
2. Molecules in gases move:
 - A) Only vibrate in place
 - B) In straight paths at high velocity
 - C) Slowly in fixed positions
 - D) In circular motion only
3. Which intermolecular force may exist in both solids and liquids?
 - A) Van der Waals forces
 - B) Nuclear forces
 - C) Magnetic forces
 - D) None of the above
4. The process of converting a solid directly into gas is called:
 - A) Melting
 - B) Sublimation
 - C) Condensation
 - D) Freezing
5. Deposition is the conversion of:
 - A) Solid → Liquid
 - B) Gas → Liquid
 - C) Gas → Solid
 - D) Liquid → Gas
6. Which of the following has the largest intermolecular spaces?
 - A) Solid
 - B) Liquid
 - C) Gas
 - D) Plasma
7. Crystalline solids are characterized by:
 - A) Random arrangement
 - B) Fixed geometrical lattice
 - C) No intermolecular forces

- D) No melting point
 - 8. Amorphous solids are often considered:
 - A) Supercooled liquids
 - B) Perfect crystals
 - C) Ionic solids
 - D) Metallic solids
 - 9. Which technique is commonly used to determine crystal structure?
 - A) UV spectroscopy
 - B) X-ray diffraction
 - C) Titration
 - D) Distillation
 - 10. Polymorphism means:
 - A) Same compound with different molecular formula
 - B) Same compound with different crystalline forms
 - C) Different compounds with same structure
 - D) Liquid-solid equilibrium
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Part B: True / False (5 Questions)

- 11. Liquids have fixed volume but not fixed shape.
 - 12. Intermolecular spaces are smallest in gases.
 - 13. Sublimation occurs without passing through the liquid phase.
 - 14. Crystalline solids are generally more stable than amorphous solids.
 - 15. Amorphous forms of drugs are usually more soluble than crystalline forms.
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Part C: Complete (5 Questions)

- 16. Conversion of liquid into solid by decreasing temperature is called _____.
 - 17. Conversion of gas into liquid is called _____.
 - 18. The fourth state of matter discussed in the lecture is _____.
 - 19. The structural arrangement of crystalline solids forms a _____.
 - 20. Heat absorbed when a solid melts is called _____.
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Part D: Short Notes (3 Questions)

21. Define sublimation.
 22. Define polymorphism.
 23. Explain latent heat of fusion.
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Part E: Compare / Explain (2 Questions)

24. Compare between crystalline solids and amorphous solids in terms of structure and stability.
 25. Explain why amorphous drugs are usually more soluble than crystalline drugs.
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Lecture 2 – Solubility (25 Questions)

Part A: MCQ (10 Questions)

1. Solubility is defined as:
 - A) Minimum amount of solute dissolved
 - B) Maximum amount of solute dissolved in given solvent
 - C) Density of a solution
 - D) Molecular weight of solute
2. A “freely soluble” substance requires approximately:
 - A) Less than 1 part solvent
 - B) 1–10 parts solvent
 - C) 30–100 parts solvent
 - D) More than 10,000 parts solvent
3. 5% w/w solution means:
 - A) 5 g solute / 100 g solution
 - B) 5 g solute / 100 mL solvent
 - C) 5 mL solute / 100 mL solution
 - D) 5 g solute / 100 mL solution
4. 10% w/v solution means:
 - A) 10 g solute in 100 mL solution
 - B) 10 mL solute in 100 mL solution
 - C) 10 g solute in 1 L solution
 - D) 10 moles solute in solution
5. Molarity is defined as:
 - A) Moles of solute / kg solvent

- B) Moles of solute / liter solution
 - C) Grams / liter
 - D) Volume / mole
6. Molality is defined as:
- A) Moles / liter solution
 - B) Moles / kg solvent
 - C) Grams / liter
 - D) Moles / mole fraction
7. Mole fraction is calculated using:
- A) Mass of solute
 - B) Volume of solute
 - C) Number of moles of component / total moles
 - D) Density
8. Which factor must be controlled during solubility determination?
- A) Temperature
 - B) Purity
 - C) Equilibrium
 - D) All of the above
9. The official method of solubility determination is:
- A) Analytical method
 - B) Synthetic method
 - C) Phase method
 - D) Optical method
10. Phase solubility analysis studies solubility with respect to:
- A) Pressure only
 - B) Temperature or pH
 - C) Molecular weight
 - D) Crystal color
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Part B: True / False (5 Questions)

- 11. Solutions can exist in solid, liquid and gaseous states.
 - 12. Air is an example of gas-gas solution.
 - 13. Impurities do not affect solubility measurements.
 - 14. Temperature should be controlled when determining solubility.
 - 15. A saturated solution must be obtained before analysis.
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Part C: Complete (5 Questions)

16. The number of milliliters of solvent required to dissolve 1 g of solute is called _____ expression of solubility.
 17. Percent concentration by mass is written as _____.
 18. Percent concentration by volume is written as _____.
 19. Number of moles per liter is called _____.
 20. Number of moles per kg solvent is called _____.
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Part D: Short Notes (3 Questions)

21. Define solubility.
 22. Write the formula of molarity.
 23. Write the formula of mole fraction.
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Part E: Compare / Explain (2 Questions)

24. Compare between molarity and molality.
 25. Explain why temperature control is important in solubility determination.
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Lecture 3 – Factors Affecting Solubility (25 Questions)

Part A: MCQ (10 Questions)

1. Which factor commonly increases the solubility of most solid drugs?
 - A) Decreasing temperature
 - B) Increasing temperature
 - C) Increasing pressure
 - D) Decreasing particle size only
2. When dissolution is endothermic, increasing temperature will:
 - A) Decrease solubility
 - B) Increase solubility
 - C) Have no effect
 - D) Stop dissolution
3. An exothermic dissolution process will generally show:

- A) Increased solubility with temperature
 - B) Decreased solubility with temperature
 - C) No change
 - D) Infinite solubility
4. Which of the following affects solubility due to crystal arrangement?
- A) Molecular weight
 - B) Polymorphism
 - C) Density
 - D) Color
5. Amorphous forms of drugs usually have:
- A) Lower solubility
 - B) Higher solubility
 - C) No solubility
 - D) Same solubility
6. Decreasing particle size generally:
- A) Decreases surface area
 - B) Increases dissolution rate
 - C) Stops dissolution
 - D) Reduces solubility completely
7. Salts of weak acids or bases are usually:
- A) Less soluble
 - B) More soluble
 - C) Insoluble
 - D) Unstable only
8. Increasing polarity of a molecule generally:
- A) Decreases water solubility
 - B) Increases water solubility
 - C) Has no effect
 - D) Stops dissolution
9. Which factor relates to interaction between drug and solvent molecules?
- A) Crystal color
 - B) Molecular structure
 - C) Container size
 - D) Tablet shape
10. pH affects solubility mainly for:
- A) Neutral compounds
 - B) Weak acids and bases
 - C) Metals
 - D) Inert gases
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Part B: True / False (5 Questions)

11. Most dissolution processes are endothermic.
 12. Increasing temperature always decreases solubility.
 13. Polymorphs are chemically identical but physically different.
 14. Amorphous forms often dissolve faster than crystalline forms.
 15. Particle size has no effect on dissolution rate.
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Part C: Complete (5 Questions)

16. Heat absorbed or released during dissolution is called _____.
 17. The ability of a compound to exist in more than one crystalline form is called _____.
 18. The random non-crystalline form of a solid is called _____.
 19. Smaller particle size increases the _____ area of a solid.
 20. The acidity or basicity of a solution is expressed as _____.
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Part D: Short Notes (3 Questions)

21. Explain the effect of temperature on solubility.
 22. Explain polymorphism in pharmaceutical compounds.
 23. Explain the effect of particle size on dissolution.
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Part E: Compare / Explain (2 Questions)

24. Compare between endothermic and exothermic dissolution processes.
 25. Explain how pH affects the solubility of weakly acidic drugs.
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Lecture 4 – Phase Equilibrium (25 Questions)

Part A: MCQ (10 Questions)

1. The phase rule was developed by:
A) Newton

- B) Gibbs
 - C) Einstein
 - D) Boyle
2. The Gibbs phase rule equation is:
- A) $F = P + C$
 - B) $F = C - P + 2$
 - C) $F = C + P$
 - D) $F = P - C + 1$
3. A phase is defined as:
- A) Any solid material
 - B) Any homogeneous part of a system
 - C) Only liquids
 - D) Only gases
4. In the system ice–water–water vapor, the number of components is:
- A) 1
 - B) 2
 - C) 3
 - D) 4
5. Degrees of freedom represent:
- A) Number of compounds
 - B) Number of independent variables
 - C) Number of phases
 - D) Molecular weight
6. When $F = 2$, the system is:
- A) Invariant
 - B) Univariant
 - C) Bivariant
 - D) Non-existent
7. When three phases coexist in equilibrium, the system is:
- A) Bivariant
 - B) Univariant
 - C) Invariant
 - D) Multivariant
8. The point where three phases coexist is called:
- A) Critical point
 - B) Triple point
 - C) Boiling point
 - D) Freezing point
9. Phase diagrams represent the relationship between:
- A) Temperature, pressure and phases

- B) Color and density
 - C) Molecular weight and volume
 - D) Density and color
10. Freeze drying is also known as:
- A) Distillation
 - B) Lyophilization
 - C) Crystallization
 - D) Filtration
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Part B: True / False (5 Questions)

- 11. A mixture of gases represents a single phase.
 - 12. Oil and water represent two phases.
 - 13. The triple point represents equilibrium between three phases.
 - 14. Freeze drying removes water by sublimation.
 - 15. Phase diagrams show the effect of temperature and pressure on phases.
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Part C: Complete (5 Questions)

- 16. The number of components in a system is represented by _____.
 - 17. The number of phases in a system is represented by _____.
 - 18. Degrees of freedom are represented by the symbol _____.
 - 19. The coexistence of solid, liquid and vapor occurs at the _____ point.
 - 20. Drying by sublimation from frozen state is called _____.
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Part D: Short Notes (3 Questions)

- 21. Define phase.
 - 22. Define component in phase rule.
 - 23. Explain degrees of freedom.
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Part E: Compare / Explain (2 Questions)

- 24. Compare between phase and component.

25. Explain the pharmaceutical importance of freeze drying.

Comprehensive Midterm Questions (100 Questions)

Part A: MCQ (50 Questions)

1. Which state of matter has particles that only vibrate in fixed positions?
A) Gas
B) Liquid
C) Solid
D) Plasma
2. The process of liquid changing to gas is called:
A) Freezing
B) Evaporation
C) Deposition
D) Condensation
3. Which solid type has ordered crystal lattice?
A) Amorphous
B) Crystalline
C) Colloidal
D) Polymeric
4. Which technique is used to study crystal structure?
A) X-ray diffraction
B) Filtration
C) Distillation
D) Centrifugation
5. Polymorphism refers to:
A) Different chemical composition
B) Same compound different crystal form
C) Different molecular weight
D) Same density
6. Which state has the highest intermolecular spaces?
A) Solid
B) Liquid
C) Gas
D) Crystal
7. Solubility is defined as:
A) Minimum amount dissolved

- B) Maximum amount dissolved in solvent
- C) Volume of solvent
- D) Density of solution

8. 5% w/v solution means:

- A) 5 g solute in 100 mL solution
- B) 5 mL solute in 100 mL solution
- C) 5 g in 1 L
- D) 5 moles

9. Molarity is:

- A) Moles per kg solvent
- B) Moles per liter solution
- C) Grams per liter
- D) Mole fraction

10. Molality is:

- A) Moles per kg solvent
- B) Moles per liter solution
- C) Volume fraction
- D) Density

11. Mole fraction equals:

- A) mass fraction
- B) moles component / total moles
- C) density ratio
- D) weight ratio

12. Increasing temperature usually:

- A) decreases solubility
- B) increases solubility
- C) stops dissolution
- D) no effect

13. Exothermic dissolution means:

- A) heat absorbed
- B) heat released
- C) no heat change
- D) melting

14. Endothermic dissolution means:

- A) heat absorbed
- B) heat released
- C) freezing
- D) condensation

15. Smaller particle size:

- A) reduces dissolution

- B) increases surface area
 - C) stops dissolution
 - D) reduces solubility
16. Salts of weak acids are usually:
- A) less soluble
 - B) more soluble
 - C) insoluble
 - D) unstable
17. Solubility of weak acids is affected by:
- A) pH
 - B) pressure
 - C) color
 - D) density
18. The phase rule equation is:
- A) $F = C - P + 2$
 - B) $F = P - C$
 - C) $F = C + P$
 - D) $F = P + 2$
19. Phase rule was proposed by:
- A) Einstein
 - B) Gibbs
 - C) Newton
 - D) Dalton
20. Phase is defined as:
- A) homogeneous part of system
 - B) compound
 - C) molecule
 - D) mixture
21. Ice-water-vapor system has components:
- A) 1
 - B) 2
 - C) 3
 - D) 4
22. Triple point means:
- A) two phases equilibrium
 - B) three phases equilibrium
 - C) one phase
 - D) four phases
23. Freeze drying is also called:
- A) filtration

- B) lyophilization
 - C) extraction
 - D) evaporation
24. Freeze drying removes water by:
- A) boiling
 - B) sublimation
 - C) condensation
 - D) melting
25. Phase diagrams represent relation between:
- A) pressure and temperature
 - B) color and mass
 - C) density and shape
 - D) size and weight
26. Liquid crystals are:
- A) purely solids
 - B) intermediate state
 - C) gases
 - D) metals
27. Amorphous solids are considered:
- A) supercooled liquids
 - B) crystals
 - C) gases
 - D) polymers
28. Latent heat of fusion is heat required to:
- A) vaporize
 - B) melt solid
 - C) freeze gas
 - D) condense liquid
29. Higher intermolecular forces produce:
- A) lower melting point
 - B) higher melting point
 - C) no melting
 - D) evaporation
30. Solubility curve relates:
- A) temperature and solubility
 - B) pressure and density
 - C) color and mass
 - D) shape and weight
31. Increasing polarity usually:
- A) increases water solubility

- B) decreases solubility
 - C) no effect
 - D) stops dissolution
32. Hydrophilic groups:
- A) increase water solubility
 - B) decrease solubility
 - C) no effect
 - D) reduce dissolution
33. Molecular surface area increase usually:
- A) increases solubility
 - B) decreases solubility
 - C) no effect
 - D) doubles solubility
34. Phase with same properties throughout is:
- A) homogeneous
 - B) heterogeneous
 - C) mixture
 - D) compound
35. Oil and water system contains:
- A) one phase
 - B) two phases
 - C) three phases
 - D) four phases
36. Saturated solution means:
- A) maximum solute dissolved
 - B) minimum solute
 - C) no solute
 - D) half solute
37. Analytical method of solubility uses:
- A) spectroscopy
 - B) filtration
 - C) evaporation
 - D) heating only
38. Synthetic method determines solubility by:
- A) dissolving known amount
 - B) cooling solution
 - C) heating solvent
 - D) filtering crystals
39. Phase solubility analysis studies:
- A) solubility vs conditions

- B) density
- C) color
- D) weight

40. Dissolution rate depends on:

- A) particle size
- B) surface area
- C) temperature
- D) all of the above

41. Increasing temperature increases solubility in:

- A) endothermic systems
- B) exothermic systems
- C) gases only
- D) metals only

42. Crystalline drugs are usually:

- A) less soluble
- B) more soluble
- C) unstable
- D) gases

43. Amorphous drugs are:

- A) higher energy
- B) lower energy
- C) neutral energy
- D) no energy

44. Stability of crystals is usually:

- A) higher
- B) lower
- C) equal
- D) none

45. The variable conditions defining equilibrium are:

- A) temperature pressure concentration
- B) color density mass
- C) shape weight size
- D) charge only

46. When $F = 1$ system is:

- A) invariant
- B) univariant
- C) bivariant
- D) multivariant

47. When $F = 0$ system is:

- A) invariant

- B) univariant
 - C) bivariant
 - D) variable
48. Condensation means:
- A) gas to liquid
 - B) liquid to gas
 - C) solid to liquid
 - D) gas to solid
49. Freezing means:
- A) liquid to solid
 - B) gas to liquid
 - C) solid to gas
 - D) vaporization
50. Melting means:
- A) solid to liquid
 - B) liquid to gas
 - C) gas to solid
 - D) condensation
-

Part B: True / False (30 Questions)

- 51. Gases have no fixed shape or volume.
- 52. Liquids have fixed volume but variable shape.
- 53. Solids have fixed shape and volume.
- 54. Sublimation occurs without liquid stage.
- 55. Amorphous forms dissolve faster.
- 56. Crystalline solids have ordered structure.
- 57. Intermolecular forces affect melting point.
- 58. Polymorphs differ in physical properties.
- 59. Solubility depends on temperature.
- 60. Impurities can affect solubility.
- 61. Saturated solution contains maximum solute.
- 62. Mole fraction depends on number of moles.
- 63. Molality uses kg solvent.
- 64. Molarity uses liter solution.
- 65. Particle size affects dissolution.
- 66. Phase rule applies to heterogeneous systems.
- 67. Triple point shows three phases equilibrium.
- 68. Oil and water are immiscible liquids.
- 69. Freeze drying uses sublimation.
- 70. Phase diagrams show equilibrium conditions.

71. Endothermic dissolution absorbs heat.
 72. Exothermic dissolution releases heat.
 73. Increasing surface area increases dissolution.
 74. Salts are often more soluble than weak acids.
 75. Polar groups increase aqueous solubility.
 76. Molecular structure affects solubility.
 77. Solvation can change crystal properties.
 78. Liquid crystals show properties of solid and liquid.
 79. Temperature affects dissolution rate.
 80. Gibbs proposed phase rule.
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Part C: Short Answer / Essay (20 Questions)

81. Define solubility.
 82. Define polymorphism.
 83. Define phase.
 84. Define component.
 85. Define degrees of freedom.
 86. Explain the process of sublimation.
 87. Explain condensation.
 88. Explain freezing.
 89. Explain melting.
 90. Explain latent heat of fusion.
 91. Explain X-ray diffraction.
 92. Explain phase rule.
 93. Explain triple point.
 94. Explain freeze drying process.
 95. Explain effect of temperature on solubility.
 96. Explain effect of particle size on dissolution.
 97. Explain role of pH in drug solubility.
 98. Explain polymorphism in pharmaceuticals.
 99. Explain difference between molarity and molality.
 100. Explain importance of solubility in drug formulation.
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Answer Key

Lecture 1

Part A: MCQ

1. Solid

2. In straight paths at high velocity
3. Van der Waals forces
4. Sublimation
5. Gas → Solid
6. Gas
7. Fixed geometrical lattice
8. Supercooled liquids
9. X-ray diffraction
10. Same compound with different crystalline forms

Part B: True / False

Part B: True / False

1. True
2. False
3. True
4. True
5. True

Part C: Complete

1. Freezing
 2. Condensation
 3. Plasma
 4. Crystal lattice
 5. Latent heat of fusion
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Lecture 2

Part A: MCQ

1. Maximum amount of solute dissolved in given solvent
2. 1–10 parts solvent

3. 5 g solute / 100 g solution
4. 10 g solute in 100 mL solution
5. Moles of solute / liter solution
6. Moles / kg solvent
7. Number of moles of component / total moles
8. All of the above
9. Analytical method
10. Temperature or pH

Part B: True / False

1. True
2. True
3. False
4. True
5. True

Part C complete

1. Maximum amount of solute that dissolves in given solvent
 2. $M = \text{moles} / \text{liter solution}$
 3. Mole fraction = moles of component / total moles
 4. Molarity uses volume of solution, Molality uses mass of solvent
 5. Because temperature strongly affects solubility equilibrium
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Lecture 3

Part A: MCQ

1. Increasing temperature
2. Increase solubility
3. Decreased solubility with temperature
4. Polymorphism
5. Higher solubility
6. Increases dissolution rate

7. More soluble
8. Increases water solubility
9. Molecular structure
10. Weak acids and bases

Part B: True / False

1. True
2. False
3. True
4. True
5. False

Part C: Complete

1. Heat of solution
2. Polymorphism
3. Amorphous solid
4. Surface area
5. pH

Lecture 4

Part A: MCQ

1. Gibbs
2. $F = C - P + 2$
3. Any homogeneous part of a system
4. 1
5. Number of independent variables
6. Bivariant
7. Invariant
8. Triple point
9. Temperature, pressure and phases
10. Lyophilization

Part B: True / False

1. True
2. True
3. True
4. True
5. True

Part C: Complete

1. C
 2. P
 3. F
 4. Triple
 5. Lyophilization
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Comprehensive Midterm

MCQ Answers

1. C
2. B
3. B
4. A
5. B
6. C
7. B
8. A
9. B
10. A
11. B
12. B
13. B
14. A
15. B
16. B
17. A
18. A

19. B
20. A
21. A
22. B
23. B
24. B
25. A
26. B
27. A
28. B
29. B
30. A
31. A
32. A
33. B
34. A
35. B
36. A
37. A
38. A
39. A
40. D
41. A
42. A
43. A
44. A
45. A
46. B
47. A
48. A
49. A
50. A

True / False Answers

51. True
52. True
53. True
54. True
55. True
56. True
57. True

- 58. True
- 59. True
- 60. True
- 61. True
- 62. True
- 63. True
- 64. True
- 65. True
- 66. True
- 67. True
- 68. True
- 69. True
- 70. True
- 71. True
- 72. True
- 73. True
- 74. True
- 75. True
- 76. True
- 77. True
- 78. True
- 79. True
- 80. True

Short Answer (Expected Points)

- 81. Maximum amount of solute that dissolves in a given solvent.
- 82. Same compound existing in different crystalline forms.
- 83. Homogeneous part of a system with uniform properties.
- 84. Minimum number of chemical constituents describing the system.
- 85. Number of independent variables needed to define equilibrium.
- 86. Direct transition of solid to gas.
- 87. Conversion of gas to liquid.
- 88. Conversion of liquid to solid.
- 89. Conversion of solid to liquid.
- 90. Heat required to melt a solid.
- 91. Technique used to determine crystal structure.
- 92. Relationship described by $F = C - P + 2$.
- 93. Point where three phases coexist in equilibrium.
- 94. Drying process by sublimation under vacuum.
- 95. Increasing temperature increases solubility in endothermic systems.
- 96. Smaller particles increase surface area and dissolution rate.

- 97. Ionization of weak acids or bases changes solubility.
- 98. Different crystal forms affect drug properties and solubility.
- 99. Molarity uses volume of solution; molality uses mass of solvent.
- 100. Solubility determines drug absorption and formulation success.